🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.AT.11

Apply Two-Variable Relationships to Solve Problems

Lesson 9-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write an equation to represent the relationship between two variable quantities and use it to solve a real problem.
Language Objective: I can justify a recommendation using the words equation, solution, and at least.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

solution | substitute | at least | predict | justify

👆 *Tap any word to see what it means and a picture.*

1. ____________ — A value of the variable that makes an equation true.

2. ____________ — To replace a variable with a number so you can compute.

3. ____________ — That amount or more — the smallest amount that still works.

4. ____________ — To make a reasonable guess about a future amount using information you already have.

5. ____________ — To explain WHY an answer or recommendation makes sense, using the mathematics.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

From which bakery would you recommend Yelina purchase cupcakes? Explain?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **solution** — A value of the variable that makes an equation true.
  solución — Un valor de la variable que hace que una ecuación sea verdadera.
- [ ] **substitute** — To replace a variable with a number so you can compute.
  sustituir — Reemplazar una variable con un número para poder calcular.
- [ ] **at least** — That amount or more — the smallest amount that still works.
  al menos — Esa cantidad o más — la cantidad mínima que todavía funciona.
- [ ] **predict** — To make a reasonable guess about a future amount using information you already have.
  predecir — Hacer una estimación razonable sobre una cantidad futura usando la información que ya tienes.
- [ ] **justify** — To explain WHY an answer or recommendation makes sense, using the mathematics.
  justificar — Explicar POR QUÉ una respuesta o recomendación tiene sentido, usando las matemáticas.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

Write an ____ for each bakery, substitute Yelina's $____ budget into each one, ____ the two results, and recommend the bakery that gives more ____ for the money.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Since h is being multiplied by 2.5 in the equation, dividing by 2.5 is the operation that undoes the multiplication and isolates h.

Evidence: A strong answer names division as the inverse of multiplication and checks the solution by substituting it back in. A weak answer just says 'because that's what you do' without the inverse-operation reasoning.

Reasoning: This evidence connects the definition of solution to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Write an ____ for each bakery, substitute Yelina's $____ budget into each one, ____ the two results, and recommend the bakery that gives more ____ for the money.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** solution
2. **Notes 2:** substitute
3. **Notes 3:** at least
4. **Notes 4:** predict
5. **Notes 5:** justify
6. **Watch for this mistake:** *Adding the rate instead of multiplying — writing 2.5 + 8 = 10.5 cars for an 8-hour shift instead of 2.5 × 8 = 20. The rate applies to EVERY hour, so the total is always rate × amount.*
7. **Try It 1:** A. No — cars come in whole numbers, so round up to the next whole car *— You cannot wash seven-tenths of a car, so the model's decimal answer must be interpreted: round up so the goal is still met.*
8. **Try It 2:** A. Either can work: $12 needs at least 42 cars washed, while $14.30 needs only about 35 — so the choice depends on how many customers they expect *— Both plans reach $500 — 42 × 12 = $504 and 35 × 14.30 = $500.50 — so the real question is whether they can attract 42 cars at the lower price, or whether the higher price might turn customers away.*